

Arithmetic and Order

- **Exercise 1 (Additive Cancellation)**

Let $a, b, c \in \mathbb{Z}$. Prove that $a + b = a + c \implies b = c$.

- **Proof 1.1**

Suppose $a, b, c \in \mathbb{Z}, a + b = a + c$. Consider the additive inverse of $-a$ of a .

$$\begin{aligned} -a + (a + b) &= -a + (a + c) \\ (-a + a) + b &= (-a + a) + c \\ 0 + b &= 0 + c \\ b &= c. \end{aligned}$$

- **Exercise 2 (If adding something changes nothing)**

Suppose $a, b \in \mathbb{Z}$ and $a + b = a$. Prove that $b = 0$.

- **Proof 2.1**

Suppose $a, b \in \mathbb{Z}$ and $a + b = a$. By properties,

$$\begin{aligned} a + 0 &= a \\ a + b &= a + 0 \\ b &= 0. \end{aligned}$$

- **Exercise 3**

Suppose $a, b \in \mathbb{Z}$ and $a + b = 0$. Prove that $b = -a$.

- **Proof 3.1**

Suppose $a, b \in \mathbb{Z}$ and $a + b = 0$. By properties,

$$\begin{aligned} a + (-a) &= 0 \\ a + b &= a + (-a) \\ b &= -a. \end{aligned}$$

- **Exercise 4**

Suppose $a, b, x, y \in \mathbb{Z}, a + x = b \wedge a + y = b$. Prove $x = y$.

- **Proof 4.1**

Suppose $a, b, x, y \in \mathbb{Z}, a + x = b, a + y = b$. Then $a + x = a + y$. By cancellation law $x = y$.

- **Exercise 5**

Suppose $e \in \mathbb{Z}$ has the property that $\forall a \in \mathbb{Z}, a + e = a$. Prove that $e = 0$.

- **Proof 5.1**

Suppose $e, a \in \mathbb{Z}, a + e = a = a + 0 \implies e = 0$.

- **Exercise 6**

Suppose $a, b, c \in \mathbb{Z}, a + b = 0 \wedge a + c = 0$. Prove that $b = c$.

- **Proof 6.1**

Suppose $a, b, c \in \mathbb{Z}, a + b = 0 \wedge a + c = 0$. Then, $a + c = a + b$. By cancellation law, $c = b$.

- **Exercise 7**

Prove that $-0 = 0$.

◦ **Proof 7.1**

$0 + (-0) = 0 = 0 + 0$. By cancellation law, $-0 = 0$.

• **Exercise 8 (Negating an inequality)**

Suppose $a, b \in \mathbb{Z}, a < b \implies -b < -a$.

◦ **Proof 8.1**

Suppose $a, b \in \mathbb{Z}, a < b$. By definition, $a < b \iff a \leq b \wedge a \neq b$. Then

$$\begin{aligned} a + (-a - b) &\leq b + (-a - b) \\ (a + (-a)) + (-b) &\leq (b + (-b)) + (-a) \\ 0 + (-b) &\leq 0 + (-a) \\ -b &\leq -a \end{aligned}$$

As $-a \neq -b \implies -b < -a$.

• **Exercise 9**

Suppose $a, b \in \mathbb{Z}, c < 0, a < b$ Prove that $ac > bc$.

◦ **Proof 9.1**

Suppose $a, b \in \mathbb{Z}, c < 0, a < b$. By definition, $a < b \iff a \leq b \wedge a \neq b$. Then

$$\begin{aligned} -c > 0 &\implies (-c)a \leq (-c)b \\ &\implies -ca \leq -cb \\ &\implies -ca < -cb \quad (\text{as } a \neq b \implies ca \neq cb \implies -ca \neq -cb) \\ &\implies ca > cb \end{aligned}$$

• **Exercise 10**

Prove that $\forall n \in \mathbb{Z}, -|n| \leq n \leq |n|$.

◦ **Proof 10.1**

Let $n \in \mathbb{Z}$. We split the proof into two cases based on the definition of absolute value.

1. **Case 1:** Suppose that $n \geq 0$.

By definition, we have that $|n| = n$.

Since $n \geq 0$, by Proposition we have $-n \leq 0$, which implies

$$-n \leq n.$$

Substituting $n = |n|$, we get $-|n| \leq n \leq |n|$.

2. **Case 2:** Suppose that $n < 0$.

By definition, we have that $|n| = -n$, which implies $-|n| = n$.

Since $n < 0$ and $|n| > 0$, we have

$$n < |n|.$$

Substituting $n = -|n|$ into the left side, we get $-|n| = n \leq |n|$.

Therefore, in both cases, the inequality $-|n| \leq n \leq |n|$ holds for every $n \in \mathbb{Z}$.

• **Exercise 11**

Prove that $\forall n, a \in \mathbb{Z}, a \geq 0, |n| \leq a \iff -a \leq n \leq a$.

◦ **Proof 11.1**

Suppose $\forall n, a \in \mathbb{Z}, a \geq 0$.

($\Rightarrow$) Suppose $|n| \leq a$. Then $-a \leq -|n| \leq n \leq |n| \leq a$.

($\Leftarrow$) Suppose $n \geq 0 \Rightarrow |n| = n \Rightarrow |n| \leq a$.

Suppose $n < 0 \Rightarrow |n| = -n \Rightarrow |n| \leq a$.

- **Exercise 12 (Triangle Inequality)**

Prove that $\forall n, m \in \mathbb{Z}, |n + m| \leq |n| + |m|$.

- **Proof 12.1**

Let $n, m \in \mathbb{Z}$. By Exercise 10, we have the following inequalities for n and m :

$$-|n| \leq n \leq |n|,$$

and

$$-|m| \leq m \leq |m|.$$

Adding these two inequalities together by Properties 2.1, we obtain

$$-(|n| + |m|) \leq n + m \leq |n| + |m|.$$

We now split the proof into two cases based on the sign of $n + m$.

1. **Case 1:** Suppose that $n + m \geq 0$.

By Definition 3, we have $|n + m| = n + m$.

From the right side of the combined inequality, we directly get

$$|n + m| \leq |n| + |m|.$$

2. **Case 2:** Suppose that $n + m < 0$.

By Definition 3, we have $|n + m| = -(n + m)$.

From the left side of the combined inequality, we have $-(|n| + |m|) \leq n + m$.

By Proposition 3.1(1), multiplying by -1 reverses the inequality direction, which gives

$$-(n + m) \leq |n| + |m|.$$

Substituting $|n + m| = -(n + m)$, we get $|n + m| \leq |n| + |m|$.

Therefore, in both cases, the inequality $|n + m| \leq |n| + |m|$ holds for all $n, m \in \mathbb{Z}$.